



LEATHER
SECTOR
SKILL
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सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT
& ENTREPRENEURSHIP



कौशल गुणवत्ता प्रगति
NCVT



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REIMAGINE FUTURE

Qualification Pack



Computerized Pattern Stitching Machine Junior Technician

QP Code: LSS/Q2202

Version: 1.0

NSQF Level: 4

Leather Sector Skill Council || GCV House, 1st Floor, No.81, Nungambakkam High Road,
Nungambakkam
Chennai – 600034 || email:ramak@leatherssc.org

Qualification Pack

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LSS/Q2202: Computerized Pattern Stitching Machine Junior Technician

Brief Job Description

The individual is to perform stitching operations with the required quality of the footwear or goods & garments using computerized stitching machine and also the person can programme the machine to execute the stitching operations.

Personal Attributes

Stitcher should perform the stitching operations with good hand, leg, and eye coordination and should have good psychomotor skills

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [LSS/N2210: Preparation of materials and Computerized stitching machine for stitching operations](#)
2. [LSS/N2211: Perform program for the stitching operations](#)
3. [LSS/N2212: Inspect jigs and fix them on the machine](#)
4. [LSS/N2213: Operate the computerized pattern stitching machine with the required quality](#)
5. [LSS/N8601: Maintain health, safety & security, a clean work area & equipment, use optimized resources at the workplace, and comply with industry & organizational Requirements](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Leather
Sub-Sector	Footwear, Goods & Garments
Occupation	Stitching (Footwear), Stitching (Goods & Garments)
Country	India
NSQF Level	4
Credits	14

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Aligned to NCO/ISCO/ISIC Code	NA
Minimum Educational Qualification & Experience	12th grade Pass with NA of experience OR Completed 2nd year of the 3-year diploma after 10 with NA of experience OR Pursuing 2nd year of 3-year regular Diploma (after 10th) with NA of experience OR 8th grade pass with 2 year NTC plus 1 year NAC plus 1 year CITS with NA of experience OR 11th grade pass with 1 Year of experience relevant field OR 10th grade pass with 2 Years of experience relevant field OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience relevant experience
Minimum Level of Education for Training in School	8th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2027
NSQC Approval Date	30/04/2024
Version	1.0
Reference code on NQR	QG-04-LT-02485-2024-V1-LSSC
NQR Version	1.0

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LSS/N2210: Preparation of materials and Computerized stitching machine for stitching operations

Description

This unit is about the preparation of raw materials and other tools and equipment required for computerized pattern stitching operations

Scope

The scope covers the following :

- Prepare the materials
- Prepare the computerized pattern-stitching machine

Elements and Performance Criteria

Prepare the materials

To be competent, the user/individual on the job must be able to:

- PC1.** check the raw materials based on the specifications like color, thickness, area, etc.,
- PC2.** handle the raw materials carefully and avoid any damages to the raw materials
- PC3.** check the materials used are free from faults
- PC4.** follow the instructions on the work order or job card
- PC5.** arrange the material as per the specification
- PC6.** store the raw materials as per specification
- PC7.** report and document the fault materials
- PC8.** ensure the work area is free from hazards and wastes

Prepare the computerized pattern-stitching machine

To be competent, the user/individual on the job must be able to:

- PC9.** examine the machine for any safety-related issues and readiness for use
- PC10.** check the air connections from compressor is connected correctly
- PC11.** check the pedal and levers are connected correctly
- PC12.** set the machine with correct needle, thread and speed as per the requirements
- PC13.** test run the machine and check the emergency stop is working perfectly

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** properties of leather and non-leather materials
- KU2.** specifications given by the raw material manufacturers
- KU3.** the difference between lock stitch and chain stitch
- KU4.** the different types of needles and threads

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- KU5.** the basic maintenance of stitching machine
- KU6.** the different parts in the stitching machines like needle, bobbin and bobbin case, needle plate, tension plate, thread winder, needle bar, pressure foot, etc.,
- KU7.** the different types of stitching machines based on functions like flatbed, post bed, cylindrical, Strobel stitching, flatlock, or overlock stitching machines
- KU8.** the various parts of computerized stitching machines like the Electronic Control Unit, Display, thread tension plate, needle bar, pedal control, emergency stop button, etc.,
- KU9.** use of jigs and fixtures at the machine
- KU10.** adjusting the speed of the stitching machine
- KU11.** the programming of computerized stitching machine
- KU12.** standards, policies followed in the workplace relevant to the employment condition
- KU13.** reporting structure to be followed for communication and coordination
- KU14.** protocols and format for reporting work-related risks and problems
- KU15.** the different PPE's in the stitching operation
- KU16.** common hazards in the work area and procedures for dealing with them
- KU17.** waste disposal procedures and guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read the manuals, health and safety instructions, memos, reports, job cards, etc
- GS2.** communicate proficiently in the local language
- GS3.** listen to the issues and comments given by the co-workers, supervisors, and managers
- GS4.** communicate effectively with co-workers, supervisors, managers, etc.,
- GS5.** fill up appropriate technical forms, process charts, and activity logs in the prescribed format of the company
- GS6.** complete the tasks efficiently and accurately within the stipulated time
- GS7.** select the optimal process and appropriate tools & equipment to complete the work efficiently
- GS8.** use reasoning skills to make proper decisions and troubleshoot concerns related to own responsibilities
- GS9.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the materials</i>	10	20	-	-
PC1. check the raw materials based on the specifications like color, thickness, area, etc.,	2	2	-	-
PC2. handle the raw materials carefully and avoid any damages to the raw materials	2	2	-	-
PC3. check the materials used are free from faults	-	4	-	-
PC4. follow the instructions on the work order or job card	2	2	-	-
PC5. arrange the material as per the specification	1	4	-	-
PC6. store the raw materials as per specification	2	2	-	-
PC7. report and document the fault materials	-	4	-	-
PC8. ensure the work area is free from hazards and wastes	1	-	-	-
<i>Prepare the computerized pattern-stitching machine</i>	10	20	-	-
PC9. examine the machine for any safety-related issues and readiness for use	2	4	-	-
PC10. check the air connections from compressor is connected correctly	2	4	-	-
PC11. check the pedal and levers are connected correctly	2	4	-	-
PC12. set the machine with correct needle, thread and speed as per the requirements	2	4	-	-
PC13. test run the machine and check the emergency stop is working perfectly	2	4	-	-
NOS Total	20	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	LSS/N2210
NOS Name	Preparation of materials and Computerized stitching machine for stitching operations
Sector	Leather
Sub-Sector	Footwear, Goods & Garments
Occupation	Stitching (Footwear), Stitching (Goods & Garments)
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024

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LSS/N2211: Perform program for the stitching operations

Description

This unit is about the performing program in a computerized pattern stitching machine and loading the external program into the machine

Scope

The scope covers the following :

- Make a program on the computerized pattern-stitching machine
- Load the external program into the machine

Elements and Performance Criteria

Make a program on the computerized pattern-stitching machine

To be competent, the user/individual on the job must be able to:

- PC1.** determine the job specifications and operational objectives by referring to job cards, approved samples, work orders, planning documentation, quality control documents, etc.,
- PC2.** check the basic stitching parameters in the sample like stitch length per CM/Inch, Needle type and size, thread type and size, speed of the machine, base materials, etc.,
- PC3.** create the program for the required stitching in the machine and save the program
- PC4.** check the sequence of the program as per the process sheet
- PC5.** Select the required speed, needle, thread for the program
- PC6.** Perform test run using the program
- PC7.** use the appropriate storage medium such as a computer hard disk, pen drive, or cloud services for saving the proven program

Load the external program into the machine

To be competent, the user/individual on the job must be able to:

- PC8.** coordinate with the relevant personnel to report and rectify incorrect and inconsistent information in job specification documents
- PC9.** get the relevant program from the CAD or Product Development department
- PC10.** transfer the program to the machine by entering it in the console or transmitting it through a wired link or a data transfer device
- PC11.** validate the stitching program is correct
- PC12.** perform test runs using the program and give feedback to the relevant department
- PC13.** follow the approved procedures to resolve problems encountered with the programming, loading and editing activities
- PC14.** carry out relevant documentation as per the organizational policy

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** properties of leather and non-leather materials
- KU2.** specifications given by the raw material manufacturers
- KU3.** the difference between lock stitch and chain stitch
- KU4.** the different types of needles and threads
- KU5.** the basic maintenance of stitching machine
- KU6.** the different parts in the stitching machines like needle, bobbin and bobbin case, needle plate, tension plate, thread winder, needle bar, pressure foot, etc.,
- KU7.** the different types of stitching machines based on functions like flatbed, post bed, cylindrical, Strobel stitching, flatlock, or overlock stitching machines
- KU8.** the various parts of computerized stitching machines like the Electronic Control Unit, Display, thread tension plate, needle bar, pedal control, emergency stop button, etc.,
- KU9.** use of jigs and fixtures at the machine
- KU10.** the programming of the computerized stitching machine and how to transfer the program files into the machine
- KU11.** standards, policies followed in the workplace relevant to the employment condition
- KU12.** reporting structure to be followed for communication and coordination
- KU13.** protocols and format for reporting work-related risks and problems
- KU14.** the different PPE's in the stitching operation
- KU15.** common hazards in the work area and procedures for dealing with them
- KU16.** waste disposal procedures and guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read the manuals, health and safety instructions, memos, reports, job cards, etc.,
- GS2.** communicate proficiently in the local language
- GS3.** listen to the issues and comments given by the co-workers, supervisors, and managers
- GS4.** communicate effectively with co-workers, supervisors, managers, etc.,
- GS5.** fill up appropriate technical forms, process charts, and activity logs in the prescribed format of the company
- GS6.** complete the tasks efficiently and accurately within the stipulated time
- GS7.** select the optimal process and appropriate tools & equipment to complete the work efficiently
- GS8.** use reasoning skills to make proper decisions and troubleshoot concerns related to own responsibilities
- GS9.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Make a program on the computerized pattern-stitching machine</i>	14	32	-	-
PC1. determine the job specifications and operational objectives by referring to job cards, approved samples, work orders, planning documentation, quality control documents, etc.,	2	4	-	-
PC2. check the basic stitching parameters in the sample like stitch length per CM/Inch, Needle type and size, thread type and size, speed of the machine, base materials, etc.,	2	4	-	-
PC3. create the program for the required stitching in the machine and save the program	2	8	-	-
PC4. check the sequence of the program as per the process sheet	2	4	-	-
PC5. Select the required speed, needle, thread for the program	2	4	-	-
PC6. Perform test run using the program	2	4	-	-
PC7. use the appropriate storage medium such as a computer hard disk, pen drive, or cloud services for saving the proven program	2	4	-	-
<i>Load the external program into the machine</i>	16	38	-	-
PC8. coordinate with the relevant personnel to report and rectify incorrect and inconsistent information in job specification documents	2	6	-	-
PC9. get the relevant program from the CAD or Product Development department	2	6	-	-
PC10. transfer the program to the machine by entering it in the console or transmitting it through a wired link or a data transfer device	2	4	-	-
PC11. validate the stitching program is correct	4	8	-	-
PC12. perform test runs using the program and give feedback to the relevant department	2	6	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. follow the approved procedures to resolve problems encountered with the programming, loading and editing activities	2	4	-	-
PC14. carry out relevant documentation as per the organizational policy	2	4	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	LSS/N2211
NOS Name	Perform program for the stitching operations
Sector	Leather
Sub-Sector	Footwear, Goods & Garments, Non-Leather Footwear
Occupation	Stitching (Footwear), Stitching (Goods & Garments)
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024

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LSS/N2212: Inspect jigs and fix them on the machine

Description

This unit is about the inspection of the jigs/fixtures for the pattern stitching machine and fixing them in the machine

Scope

The scope covers the following :

- Perform inspection of the Jigs and fixtures
- Fixing them in the pattern stitching machine

Elements and Performance Criteria

Perform inspection of the Jigs and fixtures

To be competent, the user/individual on the job must be able to:

- PC1.** select and prepare the appropriate jigs and other tools for testing with proving the program
- PC2.** Inspect the jigs as per the correct specifications
- PC3.** check the jig dimensions like length and width are matching with the dimensions of computerized stitching machine
- PC4.** inspect the closing of jigs are perfect without any tolerance
- PC5.** check the needles are mounted in positions corresponding to the program selected

Fix the machine

To be competent, the user/individual on the job must be able to:

- PC6.** check the jigs are correctly fixed in the pattern sewing machine
- PC7.** check the component is free from edge burs, correctly cut and proper thickness
- PC8.** ensure the component is as per dimension and correctly fixing in the jigs
- PC9.** ensure there is no damage to the jigs while using it
- PC10.** perform a test run with the test component on the jig and provide feedback to relevant authority

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** properties of leather and non-leather materials
- KU2.** specifications given by the raw material manufacturers
- KU3.** the difference between lock stitch and chain stitch
- KU4.** the different types of needles and threads
- KU5.** the basic maintenance of stitching machine
- KU6.** the different parts in the stitching machines like needle, bobbin and bobbin case, needle plate, tension plate, thread winder, needle bar, pressure foot, etc.,

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- KU7.** the different types of stitching machines based on functions like flatbed, post bed, cylindrical, Strobel stitching, flatlock, or overlock stitching machines
- KU8.** the various parts of computerized stitching machines like the Electronic Control Unit, Display, thread tension plate, needle bar, pedal control, emergency stop button, etc.,
- KU9.** use of jigs and fixtures at the machine
- KU10.** adjusting the speed of the stitching machine
- KU11.** the programming of computerized stitching machine
- KU12.** standards, policies followed in the workplace relevant to the employment condition
- KU13.** reporting structure to be followed for communication and coordination
- KU14.** protocols and format for reporting work-related risks and problems
- KU15.** the different PPE's in the stitching operation
- KU16.** common hazards in the work area and procedures for dealing with them
- KU17.** waste disposal procedures and guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read the manuals, health and safety instructions, memos, reports, job cards, etc.,
- GS2.** communicate proficiently in the local language
- GS3.** listen to the issues and comments given by the co-workers, supervisors, and managers
- GS4.** communicate effectively with co-workers, supervisors, managers, etc.,
- GS5.** fill up appropriate technical forms, process charts, and activity logs in the prescribed format of the company
- GS6.** complete the tasks efficiently and accurately within the stipulated time
- GS7.** select the optimal process and appropriate tools & equipment to complete the work efficiently
- GS8.** use reasoning skills to make proper decisions and troubleshoot concerns related to own responsibilities
- GS9.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform inspection of the Jigs and fixtures</i>	5	20	-	-
PC1. select and prepare the appropriate jigs and other tools for testing with proving the program	1	4	-	-
PC2. Inspect the jigs as per the correct specifications	1	6	-	-
PC3. check the jig dimensions like length and width are matching with the dimensions of computerized stitching machine	1	6	-	-
PC4. inspect the closing of jigs are perfect without any tolerance	1	2	-	-
PC5. check the needles are mounted in positions corresponding to the program selected	1	2	-	-
<i>Fix the machine</i>	5	20	-	-
PC6. check the jigs are correctly fixed in the pattern sewing machine	1	6	-	-
PC7. check the component is free from edge burs, correctly cut and proper thickness	1	5	-	-
PC8. ensure the component is as per dimension and correctly fixing in the jigs	1	4	-	-
PC9. ensure there is no damage to the jigs while using it	1	2	-	-
PC10. perform a test run with the test component on the jig and provide feedback to relevant authority	1	3	-	-
NOS Total	10	40	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	LSS/N2212
NOS Name	Inspect jigs and fix them on the machine
Sector	Leather
Sub-Sector	Footwear, Goods & Garments
Occupation	Stitching (Footwear), Stitching (Goods & Garments)
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024

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LSS/N2213: Operate the computerized pattern stitching machine with the required quality

Description

This unit describes the operation of a computerized pattern-stitching machine with the required quality

Scope

The scope covers the following :

- Operate the pattern stitching machine
- Contribute to achieving product quality in a computerized pattern stitching machine

Elements and Performance Criteria

Operate the pattern stitching machine

To be competent, the user/individual on the job must be able to:

- PC1.** check that the program is at the correct starting point
- PC2.** ensure the jigs is fixed correctly and machine is ready operate
- PC3.** load and unload component(s) using the appropriate jigs or work-holding devices
- PC4.** check the correctness of the program by conducting a test run
- PC5.** identify the abnormal conditions while operating the machine and adjust speed, thread tension, bobbin tension and programs if required
- PC6.** check the messages shown on the machine's visual display regularly and take appropriate action
- PC7.** restart the program from the correct restart point when the machine is been stopped before the completion of the program
- PC8.** carryout the stitching operation with required pace of production
- PC9.** use the relevant industry 4.0 manufacturing technology to ensure interconnectivity, automation, real-time data collection, and analysis

Contribute to achieving product quality in a computerized pattern stitching machine

To be competent, the user/individual on the job must be able to:

- PC10.** perform the quality check on incoming raw materials for color, defects, textures, strength, etc., as per the sample/specifications
- PC11.** check the quality of the stitching needle and thread in the following parameters needle number and point as per the specification sheet, needle bent, thread break, excessive thread tension, thread fraying, etc.,
- PC12.** check that the stitching is done properly as per the design specifications and that no damage happens to the appearance and quality
- PC13.** carry out the quality checks at the specified intervals on the finished product
- PC14.** check any stitch quality issues like stitch open, loop stitch, stitch break, roller impression, uneven stitching if any
- PC15.** report defects in the materials, tools, and equipment to the supervisor

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PC16. report the problems in the given format and maintain the quality record and documentation

PC17. leave work area safe and secure when work is complete

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** properties of leather and non-leather materials
- KU2.** specifications given by the raw material manufacturers
- KU3.** the difference between lock stitch and chain stitch
- KU4.** the different types of needles and threads
- KU5.** the basic maintenance of stitching machine
- KU6.** the different parts in the stitching machines like needle, bobbin and bobbin case, needle plate, tension plate, thread winder, needle bar, pressure foot, etc.,
- KU7.** the different types of stitching machines based on functions like flatbed, post bed, cylindrical, Strobel stitching, flatlock, or overlock stitching machines
- KU8.** the various parts of computerized stitching machines like the Electronic Control Unit, Display, thread tension plate, needle bar, pedal control, emergency stop button, etc.,
- KU9.** use of jigs and fixtures at the machine
- KU10.** adjusting the speed of the stitching machine
- KU11.** the programming of computerized stitching machine
- KU12.** standards, policies followed in the workplace relevant to the employment condition
- KU13.** reporting structure to be followed for communication and coordination
- KU14.** protocols and format for reporting work-related risks and problems
- KU15.** the different PPE's in the stitching operation
- KU16.** common hazards in the work area and procedures for dealing with them
- KU17.** waste disposal procedures and guidelines

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read the manuals, health and safety instructions, memos, reports, job cards, etc.,
- GS2.** communicate proficiently in the local language
- GS3.** listen to the issues and comments given by the co-workers, supervisors, and managers
- GS4.** communicate effectively with co-workers, supervisors, managers, etc.,
- GS5.** fill up appropriate technical forms, process charts, and activity logs in the prescribed format of the company
- GS6.** complete the tasks efficiently and accurately within the stipulated time
- GS7.** select the optimal process and appropriate tools & equipment to complete the work efficiently
- GS8.** use reasoning skills to make proper decisions and troubleshoot concerns related to own responsibilities
- GS9.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Operate the pattern stitching machine</i>	20	48	-	-
PC1. check that the program is at the correct starting point	4	6	-	-
PC2. ensure the jigs is fixed correctly and machine is ready operate	2	4	-	-
PC3. load and unload component(s) using the appropriate jigs or work-holding devices	3	6	-	-
PC4. check the correctness of the program by conducting a test run	2	6	-	-
PC5. identify the abnormal conditions while operating the machine and adjust speed, thread tension, bobbin tension and programs if required	2	6	-	-
PC6. check the messages shown on the machine's visual display regularly and take appropriate action	2	6	-	-
PC7. restart the program from the correct restart point when the machine is been stopped before the completion of the program	2	6	-	-
PC8. carryout the stitching operation with required pace of production	1	4	-	-
PC9. us the relevant industry 4.0 manufacturing technology to ensure interconnectivity, automation, real-time data collection, and analysis	2	4	-	-
<i>Contribute to achieving product quality in a computerized pattern stitching machine</i>	10	32	-	-
PC10. perform the quality check on incoming raw materials for color, defects, textures, strength, etc., as per the sample/specifications	2	4	-	-
PC11. check the quality of the stitching needle and thread in the following parameters needle number and point as per the specification sheet, needle bent, thread break, excessive thread tension, thread fraying, etc.,	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check that the stitching is done properly as per the design specifications and that no damage happens to the appearance and quality	2	4	-	-
PC13. carry out the quality checks at the specified intervals on the finished product	2	4	-	-
PC14. check any stitch quality issues like stitch open, loop stitch, stitch break, roller impression, uneven stitching if any	2	4	-	-
PC15. report defects in the materials, tools, and equipment to the supervisor	-	4	-	-
PC16. report the problems in the given format and maintain the quality record and documentation	-	4	-	-
PC17. leave work area safe and secure when work is complete	-	4	-	-
NOS Total	30	80	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	LSS/N2213
NOS Name	Operate the computerized pattern stitching machine with the required quality
Sector	Leather
Sub-Sector	Footwear, Goods & Garments
Occupation	Stitching (Footwear), Stitching (Goods & Garments)
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/04/2024
Next Review Date	30/04/2027
NSQC Clearance Date	30/04/2024

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LSS/N8601: Maintain health, safety & security, a clean work area & equipment, use optimized resources at the workplace, and comply with industry & organizational Requirements

Description

This unit is about maintaining good hygiene, health, and workplace safety standards, maintaining work areas and activities to ensure tools and machines are maintained as per norms, understanding and complying with the requirements of the organization and respecting everyone at the workplace, adopting sustainable practices and optimizing the use of resources, especially material, energy, and waste, in day-to-day operations at the workplace.

Scope

The scope covers the following :

- Compliance with health, safety, and security requirements
- Maintain hygiene and sanitation
- Maintain the work area
- Perform maintenance operations
- Compliance with industry, regulatory and organizational requirements
- Respect diversity
- Material conservation practices
- Energy/electricity conservation practices
- Effective waste management/recycling practices

Elements and Performance Criteria

Compliance with health, safety and security requirements at the workplace

To be competent, the user/individual on the job must be able to:

- PC1.** comply with health and safety-related instructions applicable to the workplace
- PC2.** use the personal protective equipment as per protocol
- PC3.** maintain a healthy lifestyle and guard against dependency on intoxicants
- PC4.** follow environment management system related procedures
- PC5.** monitor the workplace and work processes for potential risks and threats
- PC6.** carry out periodic walk-through to keep the work area free from hazards and obstructions, if assigned
- PC7.** report hazards and potential risks/ threats to supervisors or other authorized personnel
- PC8.** participate in mock drills/ evacuation procedures organized at the workplace
- PC9.** undertake first aid training, fire-fighting drill, and emergency response training, etc.,
- PC10.** take action based on instructions in the event of fire, emergencies or accidents
- PC11.** safely handle and move waste and debris
- PC12.** follow organization procedures for shutdown and evacuation when require

Maintain hygiene and sanitation at the workplace

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To be competent, the user/individual on the job must be able to:

PC13. maintain personal body hygiene and dental hygiene

PC14. follow social distancing at the workplace

PC15. ensure not to go to work if unwell, to avoid the risk of spreading an infection to other people

PC16. cover the mouth with an elbow/handkerchief or turn away from people while sneezing or coughing

PC17. ensure equipment, work area, restrooms, etc. are sanitized before and after sessions

PC18. ensure regular vaccinations to avoid any spreading of diseases

Maintain the work area

To be competent, the user/individual on the job must be able to:

PC19. handle materials, machinery, equipment and tools safely and correctly

PC20. use correct lifting and handling procedures

PC21. use materials to minimize waste

Perform maintenance operations

To be competent, the user/individual on the job must be able to:

PC22. Perform the cleaning of machine or tools at the end of each shift

PC23. carry out running maintenance within agreed schedules

PC24. report unsafe equipment and other dangerous occurrences

PC25. carry out works in a comfortable position with the correct posture

PC26. maintain proper lighting, and ventilation to make sure general comfort is there while working

Compliance with industry, regulatory and organizational requirements

To be competent, the user/individual on the job must be able to:

PC27. carry out work functions in accordance with legislation and regulations, organizational guidelines, and procedures

PC28. seek and obtain clarifications on policies and procedures from the authorized personnel

PC29. follow the policies and procedures within the work practices

PC30. identify and report any possible deviation to these requirements

Respect diversity

To be competent, the user/individual on the job must be able to:

PC31. transact with everyone without any personal bias like gender, disability, caste, religion, color, sexual orientation, and culture

PC32. identify the indicators of harassment and discrimination

PC33. report incidents of harassment and discrimination to appropriate authority

PC34. respect the personal and professional space of others

Material conservation practices

To be competent, the user/individual on the job must be able to:

PC35. identify ways to optimize the usage of material including water in various activities /processes

PC36. check for spills/leakages in various processes and plug the spills and leakages

PC37. use the optimized amount of chemicals that have petroleum products (adhesives, leather chemicals, chrome salt, cleaning agents, IPA, etc.)

Qualification Pack

PC38. use proper PPE when in contact with the above materials as per specifications

Energy/electricity conservation practices

To be competent, the user/individual on the job must be able to:

PC39. identify ways to optimize the usage of electricity/energy in various activities/processes

PC40. check if the equipment/machine is generally functioning before commencing work and rectify wherever required

PC41. ensure electrical equipment and appliances are properly connected and turned off when not in use

Effective waste management/recycling practices

To be competent, the user/individual on the job must be able to:

PC42. identify recyclable and non-recyclable, and hazardous waste generated

PC43. segregate waste into different categories

PC44. dispose of non-recyclable waste appropriately

PC45. deposit recyclable and reusable material at the identified location

PC46. follow processes specified for disposal of hazardous waste

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the health and safety-related practices applicable at the workplace

KU2. the potential hazards, risks, and threats based on the nature of operations

KU3. the common hazards in the work area and procedure for dealing with them

KU4. the organizational procedures for safe handling of equipment and machine operations

KU5. details of personal protective equipment and method to use it

KU6. the signage used for personnel protective equipment, fire and safety, and first aid

KU7. layout of the plant and details of emergency exits, escape routes, emergency equipment, and assembly points

KU8. details of personnel trained in first aid, fire-fighting, and emergency response

KU9. actions to take in the event of mock drills/ evacuation procedures or actual accident, emergency, or fire

KU10. the storage and disposal of hazardous substances and wastes

KU11. relevance of health and safety requirements applicable at the workplace

KU12. importance of sound health, hygiene, and good habits

KU13. the ill-effects of alcohol, tobacco, and other drugs

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read the manuals, health and safety instructions, memos, reports, job cards, etc.

GS2. communicate proficiently in the local language

GS3. listen the issues and comments given by the co-workers, supervisors, and managers

Qualification Pack

- GS4.** communicate effectively with co-workers, supervisors, managers, etc.,
- GS5.** fill up appropriate technical forms, process charts, activity logs in the prescribed format of the company
- GS6.** analyze the defects and the procedure for dealing with it
- GS7.** complete the tasks efficiently and accurately within the stipulated time
- GS8.** organize the material requirements sequentially
- GS9.** select the optimal process and appropriate tools & equipment to complete the work efficiently
- GS10.** use reasoning skills to make proper decisions and troubleshoot concerns related to own responsibilities
- GS11.** use problem-solving skills to solve the problems that occurred during relevant operation

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Compliance with health, safety and security requirements at the workplace</i>	6	24	-	-
PC1. comply with health and safety-related instructions applicable to the workplace	1	2	-	-
PC2. use the personal protective equipment as per protocol	1	2	-	-
PC3. maintain a healthy lifestyle and guard against dependency on intoxicants	-	2	-	-
PC4. follow environment management system related procedures	-	2	-	-
PC5. monitor the workplace and work processes for potential risks and threats	1	2	-	-
PC6. carry out periodic walk-through to keep the work area free from hazards and obstructions, if assigned	-	2	-	-
PC7. report hazards and potential risks/ threats to supervisors or other authorized personnel	-	3	-	-
PC8. participate in mock drills/ evacuation procedures organized at the workplace	-	3	-	-
PC9. undertake first aid training, fire-fighting drill, and emergency response training, etc.,	1	3	-	-
PC10. take action based on instructions in the event of fire, emergencies or accidents	1	-	-	-
PC11. safely handle and move waste and debris	-	2	-	-
PC12. follow organization procedures for shutdown and evacuation when require	1	1	-	-
<i>Maintain hygiene and sanitation at the workplace</i>	4	16	-	-
PC13. maintain personal body hygiene and dental hygiene	1	2	-	-
PC14. follow social distancing at the workplace	-	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. ensure not to go to work if unwell, to avoid the risk of spreading an infection to other people	1	3	-	-
PC16. cover the mouth with an elbow/handkerchief or turn away from people while sneezing or coughing	1	3	-	-
PC17. ensure equipment, work area, restrooms, etc. are sanitized before and after sessions	-	3	-	-
PC18. ensure regular vaccinations to avoid any spreading of diseases	1	2	-	-
<i>Maintain the work area</i>	4	12	-	-
PC19. handle materials, machinery, equipment and tools safely and correctly	1	4	-	-
PC20. use correct lifting and handling procedures	2	4	-	-
PC21. use materials to minimize waste	1	4	-	-
<i>Perform maintenance operations</i>	6	18	-	-
PC22. Perform the cleaning of machine or tools at the end of each shift	1	2	-	-
PC23. carry out running maintenance within agreed schedules	1	4	-	-
PC24. report unsafe equipment and other dangerous occurrences	2	4	-	-
PC25. carry out works in a comfortable position with the correct posture	1	4	-	-
PC26. maintain proper lighting, and ventilation to make sure general comfort is there while working	1	4	-	-
<i>Compliance with industry, regulatory and organizational requirements</i>	5	10	-	-
PC27. carry out work functions in accordance with legislation and regulations, organizational guidelines, and procedures	1	2	-	-
PC28. seek and obtain clarifications on policies and procedures from the authorized personnel	2	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC29. follow the policies and procedures within the work practices	1	2	-	-
PC30. identify and report any possible deviation to these requirements	1	2	-	-
<i>Respect diversity</i>	5	10	-	-
PC31. transact with everyone without any personal bias like gender, disability, caste, religion, color, sexual orientation, and culture	2	4	-	-
PC32. identify the indicators of harassment and discrimination	1	2	-	-
PC33. report incidents of harassment and discrimination to appropriate authority	1	2	-	-
PC34. respect the personal and professional space of others	1	2	-	-
<i>Material conservation practices</i>	2	8	-	-
PC35. identify ways to optimize the usage of material including water in various activities /processes	1	2	-	-
PC36. check for spills/leakages in various processes and plug the spills and leakages	-	2	-	-
PC37. use the optimized amount of chemicals that have petroleum products (adhesives, leather chemicals, chrome salt, cleaning agents, IPA, etc.)	1	2	-	-
PC38. use proper PPE when in contact with the above materials as per specifications	-	2	-	-
<i>Energy/electricity conservation practices</i>	2	8	-	-
PC39. identify ways to optimize the usage of electricity/energy in various activities/processes	1	2	-	-
PC40. check if the equipment/machine is generally functioning before commencing work and rectify wherever required	-	4	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC41. ensure electrical equipment and appliances are properly connected and turned off when not in use	1	2	-	-
<i>Effective waste management/recycling practices</i>	6	4	-	-
PC42. identify recyclable and non-recyclable, and hazardous waste generated	2	1	-	-
PC43. segregate waste into different categories	1	1	-	-
PC44. dispose of non-recyclable waste appropriately	1	-	-	-
PC45. deposit recyclable and reusable material at the identified location	1	1	-	-
PC46. follow processes specified for disposal of hazardous waste	1	1	-	-
NOS Total	40	110	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	LSS/N8601
NOS Name	Maintain health, safety & security, a clean work area & equipment, use optimized resources at the workplace, and comply with industry & organizational Requirements
Sector	Leather
Sub-Sector	Finished Leather, Footwear, Goods & Garments, Goods and Garments, Non-Leather Footwear, Footwear/Goods & Garments, Bags and Luggage
Occupation	All Occupations
NSQF Level	3
Credits	1
Version	2.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

Qualification Pack

- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per the assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practicals for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
LSS/N2210.Preparation of materials and Computerized stitching machine for stitching operations	20	40	-	-	60	15
LSS/N2211.Perform program for the stitching operations	30	70	-	-	100	20
LSS/N2212.Inspect jigs and fix them on the machine	10	40	-	-	50	15
LSS/N2213.Operate the computerized pattern stitching machine with the required quality	30	80	-	-	110	30
LSS/N8601.Maintain health, safety & security, a clean work area & equipment, use optimized resources at the workplace, and comply with industry & organizational Requirements	40	110	-	-	150	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	150	370	-	-	520	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PPE	Personal Protective Equipment
PPE	Personal Protective Equipment

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.